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MIL-PRF-46167D
w/AMENDMENT 1
20 July 2017
SUPERSEDING
MIL-PRF-46167D
27 January 2005

PERFORMANCE SPECIFICATION

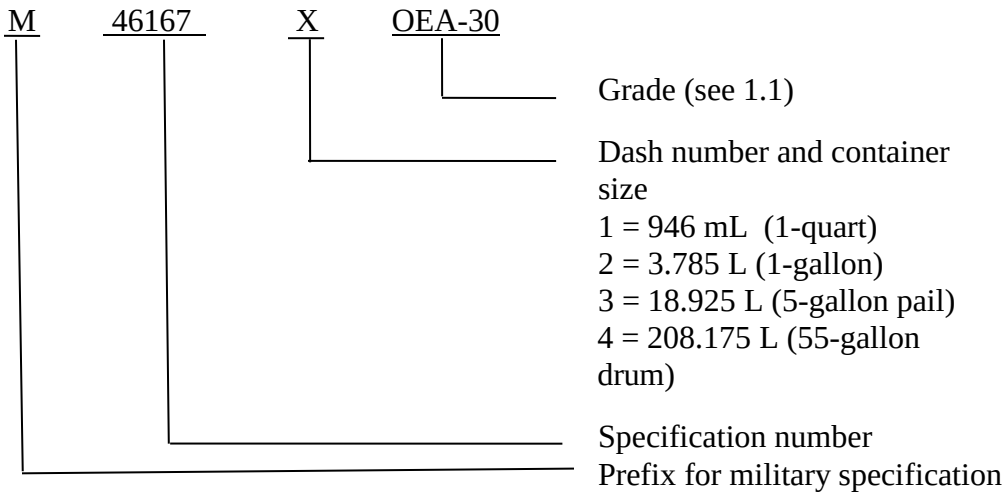
LUBRICATING OIL, INTERNAL COMBUSTION ENGINE, ARCTIC

This specification is approved for use by all Departments and agencies of the Department of Defense (DoD).

1. SCOPE

1.1 Scope. This performance specification covers one grade of internal combustion compression-ignition engine lubricating oil with military symbol OEA-30 and NATO Code O-183, suitable for arctic use (see 6.1, 6.6, and 6.9).

1.2 Part or identifying number (PIN) configuration. The PIN to be used for oils acquired to this specification are created as follows:



Comments, suggestions, or questions on this document should be addressed to Tank-automotive and Armaments Command, 6501 E. 11 Mile Road, Warren, MI 48397-5000 or emailed to usarmy.detroit.rdecom.mail.tardec-standardization@mail.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 9150

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirement documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARDS

FED-STD-791	- Lubricants, Liquid Fuels and Related Products; Methods of Testing (see 4.3.1)

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-2104	- Lubricating Oil, Internal Combustion Engine, Combat/Tactical Service
MIL-PRF-21260	- Lubricating Oil, Internal Combustion Engine, Preservative Break-in

(Copies of these documents are available online at http://quicksearch.dla.mil).

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

U.S. DEPARTMENT OF LABOR (DOL)

OSHA 29 CFR 1910.1200	- Hazard Communication; Interpretation Regarding Lubricating Oils

(Copies of this document is available online at https://www.osha.gov/pls/oshaweb/owadisp.show_document?p_table=standards&p_id=10099).

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM D92	- Standard Test Method for Flash and Fire Points by Cleveland Open Cup Tester (DoD Adopted)
ASTM D97	- Standard Test Method for Pour Point of Petroleum Oils (DoD Adopted)
ASTM D130	- Standard Test Method for Corrosiveness to Copper from Petroleum Products by Copper Strip Test (DoD Adopted)
ASTM D287	- Standard Test Method for API Gravity of Crude Petroleum and Petroleum Products (Hydrometer Method) (DoD Adopted)
ASTM D445	- Standard Test Method for Kinematic Viscosity of Transparent and Opaque Liquids (and the Calculation of Dynamic Viscosity) (DoD Adopted)
ASTM D524	- Standard Test Method for Ramsbottom Carbon Residue of Petroleum Products (DoD Adopted)
ASTM D664	- Standard Test Method for Acid Number of Petroleum Products by Potentiometric Titration (DoD Adopted)
ASTM D808	- Standard Test Method for Chlorine in New and Used Petroleum Products (High Pressure Decomposition Device Method) (DoD Adopted)
ASTM D874	- Standard Test Method for Sulfated Ash from Lubricating Oils and Additives (DoD Adopted)
ASTM D892	- Standard Test Method for Foaming Characteristics of Lubricating Oils (DoD Adopted)
ASTM D1317	- DELETED
ASTM D1500	- Standard Test Method for ASTM Color of Petroleum Products (ASTM Color Scale) (DoD Adopted)
ASTM D2270	- Standard Practice for Calculating Viscosity Index from Kinematic Viscosity at 40°C and 100°C (DoD Adopted)
ASTM D2622	- Standard Test Method for Sulfur in Petroleum Products by Wavelength Dispersive X-ray Fluorescence Spectrometry (DoD Adopted)
ASTM D2887	- Standard Test Method for Boiling Range Distribution of Petroleum Fractions by Gas Chromatography (DoD Adopted)
ASTM D2896	- Standard Test Method for Base Number of Petroleum Products by Potentiometric Perchloric Acid Titration (DoD Adopted)
ASTM D3228	- Standard Test Method for Total Nitrogen in Lubricating Oils and Fuel Oils by Modified Kjeldahl Method (DoD Adopted)

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ASTM D4057	- Standard Practice for Manual Sampling of Petroleum and Petroleum Products (DoD Adopted)
ASTM D4177	- Standard Practice for Automatic Sampling of Petroleum and Petroleum Products (DoD Adopted)
ASTM D4485	- Standard Specification for Performance of Active API Service Category Engine Oils (DoD Adopted)
ASTM D4628	- Standard Test Method for Analysis of Barium, Calcium, Magnesium, and Zinc in Unused Lubricating Oils by Atomic Absorption Spectrometry (DoD Adopted)
ASTM D4629	- Standard Test Method for Trace Nitrogen in Liquid Petroleum Hydrocarbons by Syringe/Inlet Oxidative Combustion and Chemiluminescence Detection (DoD Adopted)
ASTM D4683	- Standard Test Method for Measuring Viscosity of New and Used Engine Oils High Shear Rate and High Temperature by Tapered Bearing Simulator Viscometer at 150°C (DoD Adopted)
ASTM D4684	- Standard Test Method for Determination of Yield Stress and Apparent Viscosity of Engine Oils at Low Temperature (DoD Adopted)
ASTM D4739	- Standard Test Method for Base Number Determination by Potentiometric Hydrochloric Acid Titration (DoD Adopted)
ASTM D4741	- Standard Test Method for Measuring Viscosity at High Temperature and High Shear Rate by Tapered-Plug Viscometer (DoD Adopted)
ASTM D4927	- Standard Test Methods for Elemental Analysis of Lubricant and Additive Components - Barium, Calcium, Phosphorus, Sulfur, and Zinc by Wavelength-Dispersive X-Ray Fluorescence Spectroscopy (DoD Adopted)
ASTM D4951	- Standard Test Method for Determination of Additive Elements in Lubricating Oils by Inductively Coupled Plasma Atomic Emission Spectrometry (DoD Adopted)
ASTM D5293	- Standard Test Method for Apparent Viscosity of Engine Oils and Base Stocks between -10°C and -35°C Using Cold Cranking Simulator
ASTM D5481	- Standard Test Method for Measuring Apparent Viscosity at High Temperature and High-Shear Rate by Multi-cell Capillary Viscometer
ASTM D5800	- Standard Test Method of Evaporation Loss of Lubricating Oils by the Noack Method
ASTM D5862	- DELETED
ASTM D5966	- Standard Test Method for Evaluation of Engine Oils for Roller Follower Wear in Light-Duty Diesel Engine
ASTM D5967	- Standard Test Method for Evaluation of Diesel Engine Oils in T-8 Diesel Engine
ASTM D5968	- Standard Test Method for Evaluation of Corrosiveness of

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	Diesel Engine Oil at 121°C
ASTM D6618	- Standard Test Method for Evaluation of Engine Oils in Diesel Four-Stroke Cycle Supercharged 1M-PC Single Cylinder Oil Test Engine
ASTM D6709	- Standard Test Method for the Evaluation of Automotive Engine Oils in Sequence VIII Spark-Ignition Engine (CLR Oil Test Engine)
ASTM D6750	- Standard Test Methods for Evaluation of Engine Oils in a High-Speed, Single-Cylinder Diesel Engine 1K Procedure (0.4% Fuel Sulfur) and 1N Procedure (0.04% Fuel Sulfur)
ASTM D6894	- Standard Test Method for Evaluation of Aeration Resistance of Engine Oils in Direct-Injected Turbocharged Automotive Diesel Engine
ASTM D6922	- Standard Test Method for Determination of Homogeneity and Miscibility in Automotive Engine Oils
ASTM D6984	- Standard Test Method for Evaluation of Automotive Engine Oils in the SEQUENCE IIIF, Spark-Ignition Engine

(Copies of these documents are available online at https://www.astm.org/).

ASTM Adjunct Publications - DELETED

ALLISON TRANSMISSION DIVISION (ATD)

C-4 Fluid Specification (TES-228)

(Application for copies should be addressed to EG&G Stationary Testing, ATTN: ATF/Specialty Lab (C-4), 5904 Bandera Road, San Antonio, TX 78283-1993, or online at http://www.allisontransmission.com).

CATERPILLAR INC., ENGINE DIVISION (CAT)

Caterpillar TO-4, Fluid Requirements, VC 70

(Copies of this document are available online at https://parts.cat.com/wcs-static/pdfs/TO4+06-23-06.pdf).

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. The engine lubricating oil furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.1.1 and 6.3).

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3.2 Design, materials, manufacturing processes. Unless otherwise specified herein, the design, materials, and manufacturing process selection is the prerogative of the contractor as long as all articles submitted to the government fully meet the operating, interface, support and ownership, and operating environment requirements specified.

3.2.1 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Operating requirements. The engine lubricating oil shall meet all the finished oil and operating requirements as specified in 3.3.1 through 3.3.9.2.

3.3.1 Finished oil properties. The engine lubricating oil properties, after the application of tolerances (see 4.1.1.1), shall not exceed the maximum (max.) or fall below the minimum (min.) limits as specified in Table I.

TABLE I. Finished oil properties.

Property	Requirement
Kinematic viscosity, Centistokes (cSt):	
@ 100 degrees Celsius (°C), min.	9.3
@ 40°C	X 1/
@ -40°C, max.	18 000
@ -48°C, max.	55 000
High-temperature/high shear viscosity, Centipoise (cP), min.	2.9
Low Temperature Cranking Viscosity, cP, max. @ -35°C	6200
Low Temperature Pumping Viscosity, @ -40°C, cP,	
max.(no yield stress)	60 000
Viscosity index	X
Pour point, °C, max.	-55
Stable pour point, °C, max. 2/	-55
Flash point, °C, min.	220
Evaporative loss, percent (%), max. 3/	18
Other properties:	
Gravity	X
Carbon residue	X
Sulfur	X
Sulfated ash	X
Total acid number (TAN)	X
Total base number (TBN)	X
Phosphorus	X
Nitrogen	X
Metallic components	X
Chlorine	X
Color	X
Boiling range	X

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1/ Limits for the values marked by "X" cannot be established, as different additive technologies and base stocks will give different results. However, knowledge of these values is needed to judge performance considerations and to ascertain the potential effects of changes in oil formulations.

2/ After being cooled down to its pour point, the oil shall regain its homogeneity on standing at a temperature not more than 6°C above the pour point. However, it should not exceed the indicated limits.

3/ The indicated limits apply when ASTM D5800 is used.

3.3.2 Foaming. The oil shall not produce foam which will cause inadequate lubrication, cavitation, overflow loss of lubricant, mechanical failure, or any other detrimental effect that compromises performance.

3.3.3 Oxidation and wear protection characteristics. The oil shall protect internal loaded engine components against excessive wear and oxidation.

3.3.4 Rolling valvetrain wear protection. The oil shall protect hydraulic roller followers from excessive wear.

3.3.5 Ring-sticking, wear, and accumulation of deposits. The oil shall prevent the sticking of piston rings and port clogging, and shall minimize the wear of cylinders, rings, and loaded engine components such as cam followers, valve rocker arms, rocker arm shafts, and the oil pump and fuel injection pump drive gears.

3.3.6 Friction retention characteristics and wear. The oil shall maintain a stable coefficient of friction and shall minimize surface distress and wear during use in power shift transmissions and other cooled friction compartments, or hydraulic systems such as steering and disconnect clutches.

3.3.7 Seal deterioration. The oil shall minimize the deterioration of seal and friction materials.

3.3.8 Soot dispersing characteristics. The oil shall minimize engine deposits and increased oil viscosity by stably dispersing soot, produced during diesel combustion, in the lubricating oil.

3.3.9 Corrosion. The oil shall meet the requirements as specified in 3.3.9.1 through 3.3.9.2.

3.3.9.1 Bearing corrosion. The oil shall not corrode alloy bearings.

3.3.9.2 Metal corrosion. The oil shall not corrode metals.

3.4 Interface requirements. The oil shall meet the following interface requirements.

3.4.1 Homogeneity and miscibility. The oil shall be and shall remain homogeneous under normal operational and storage conditions.

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3.5 Support and ownership requirements. The oil shall meet the following support and ownership requirement.

3.5.1 Hazardous materials. The oil shall not contain carcinogenic or potentially carcinogenic constituents as defined under the Hazard Communication Standard 29 CFR 1910.1200.

4. VERIFICATION

4.1 Classification of inspections. Inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.1.1).
b. Conformance inspection (see 4.1.2).

4.1.1 Qualification inspection. Qualification inspection shall consist of all the verifications listed in Table II and shall be in accordance with acceptable tolerances (see 4.1.1.1).

4.1.1.1 Tolerances. Acceptable tolerances for values or limits in 3.3.1 through 3.3.9.2 shall be as specified by the qualifying activity (see 6.3).

TABLE II. Verification methods.

Title	Require-ments	Verifi-cation	Title	Require-ments	Verifi-cation
Operating requirements	3.3	4.3.3	Corrosion	3.3.9	4.3.3.9
Finished oil properties	3.3.1	4.3.3.1	Bearing corrosion	3.3.9.1	4.3.3.9.1
Foaming	3.3.2	4.3.3.2	Metal corrosion	3.3.9.2	4.3.3.9.2
Oxidation/wear protection	3.3.3	4.3.3.3	**Interface requirements**	3.4	4.3.4
Rolling valvetrain wear protection	3.3.4	4.3.3.4	Homogeneity	3.4.1	4.3.4.1
Ring sticking/wear	3.3.5	4.3.3.5	**Support and ownership requirements**	3.5	4.3.5
Friction retention/wear	3.3.6	4.3.3.6	Hazardous materials	3.5.1	4.3.5.1
Seal deterioration	3.3.7	4.3.3.7			
Soot dispersing characteristics	3.3.8	4.3.3.8			

4.1.2 Conformance inspection. The engine lubricating oil supplied under contract shall have the same base stocks and additive components, at the appropriate concentrations, as when qualified. Unless otherwise specified (see 6.2), the finished oil properties and the foaming characteristics shall be tested as specified in Table II to ensure that purchased oils are of the same composition and quality as the respective qualification samples, and to identify the products. The results obtained shall fall within the tolerances/specific values determined at the time of qualification (see 4.1.1.1).

4.2 Sampling. Samples from bulk or packaged lots (see 6.7), shall be taken for testing in accordance with ASTM D4057 or D4177, as appropriate.

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4.3 Verification methods. The types of verification methods included in this section are visual inspection, measurement, sample tests, full-scale demonstration tests, simulation, modeling, engineering evaluation, component properties analysis, and similarity to previously approved or previous qualified designs.

4.3.1 Verification alternatives. The manufacturer may propose alternative test methods, techniques, or equipment, including the application of statistical process control, tool control, or cost effective sampling procedures, to verify performance. See the contract for alternatives that replace verification required by this specification.

4.3.2 Inspection conditions. Tests may be conducted, in any sequence, in any plant or laboratory approved by the qualifying activity (see 6.2 and 6.3).

4.3.3 Operating requirements verification. Complete each test under 4.3.3.

4.3.3.1 Finished oil property verifications. Use the test methods as specified in Table III to determine that finished oil properties are as specified.

TABLE III. Finished oil property test methods.

Property	FED-STD-791 or equivalent (see 4.3.1)	ASTM
Viscosity, kinematic		D445
Viscosity, low temperature cranking		D5293
High temperature/shear		D4683, D4741, D5481
Viscosity index		D2270
Pour point		D97
Stable pour point	203	
Viscosity, low temperature, pumping		D4684
Flash point		D92
Evaporative loss		D5800
Gravity, API		D287
Color		D1500
Carbon residue		D524
Chlorine		D808
TAN		D664
TBN		D2896, D4739
Phosphorus		D4951
Sulfur		D2622, D4951
Nitrogen		D3228, D4629
Boiling range distribution		D2887
Sulfated ash		D874
Metallic components		D4628, D4927, D4951

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4.3.3.2 Foaming test. All grades of oil shall be subject to the following tests and shall exhibit the correlative results:

a. Initial test. Test oil in accordance with sequence I of ASTM D892. Not more than 10 milliliters (mL) of foam shall remain immediately following the end of the 5 minute blowing period. No foam shall remain at the end of the 10 minute settling period.
b. Intermediate test. Test oil in accordance with sequence II of ASTM D892. Not more than 20 mL of foam shall remain immediately following the end of the 5 minute blowing period. No foam shall remain at the end of the 10 minute settling period. Option A is not allowed.
c. Final test. Repeat procedures under initial test, sequence III of ASTM D892.
d. Engine Oil Aeration Test. Test Oil in accordance with ASTM D6894. No more than a 10% increase in aerated oil volume shall be present after a 30 minute settling period.

4.3.3.3 Oxidation and wear protection test. The oil shall be tested in accordance with ASTM D6984, Appendix X1 (Seq. IIIFHD Test Procedure) and shall exhibit results (single or average) meeting the following criteria:

60 h viscosity at 40°C, increase from	1 Test	2 Tests	3 Tests
10 min sample, % max	325	349	360

4.3.3.4 Rolling valvetrain wear test. The oil shall be tested in accordance with ASTM D5966 and shall exhibit results meeting the following criteria:

	1 Test	2 Tests	3 Tests
Pin wear, µm avg. max.	11.4	12.4	12.7
mils, avg. max.	0.45	0.49	0.50

4.3.3.5 Ring-sticking, wear, and accumulation of deposits tests. Test oil as specified in 4.3.3.5.1 through 4.3.3.5.2.

4.3.3.5.1 Four-stroke cycle diesel engine - low sulfur fuel. The oil shall be tested in accordance with ASTM D6750 (CAT 1N) test procedure using low-sulfur fuel. (Only one test is required; however, the test limits are adjusted according to the number of tests submitted and equivalency to original limits.) The test results shall meet the following criteria:

	1 Test	2 Tests	3 Tests
Top groove filling, % avg. max.	20	23	25
WTD, demerits, avg. max.	286.2	311.7	323.0
Top land heavy carbon, % avg. max.	3	4	5
Oil consumption, grams per kilowatt hour, (g/kW-h)	0.5	0.5	0.5
Scuffing, piston/rings/liners, avg. max.	None	None	None
Stuck rings, avg. max.	None	None	None

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4.3.3.5.2 Four-stroke cycle diesel engine - higher sulfur fuel. The oil shall be tested in accordance with ASTM D6618 (CAT 1M-PC) test procedure using high-sulfur fuel. (Only one test is required; however, when three or more tests are run, one test may be discarded and the average determined from the remaining test results.) The test results shall meet the following criteria:

Top groove filling, % avg. max.	70
WTD, avg. max.	240
Ring side clearance loss, millimeter max.	0.013
Piston ring sticking	None
Scuffing, piston/rings/liners	None

4.3.3.5.3 Two-stroke cycle diesel engine. DELETED

4.3.3.6 Friction retention and wear tests. Complete each test under 4.3.3.6.

4.3.3.6.1 Slip time and torque tests. The oil shall be tested in accordance with ATD C-4 fluid specification. The test results shall meet the following nominal criteria, as adjusted to accommodate slight changes in individual friction plate batches:

	Graphite 0 – 5,500 cycles	Paper 0 – 10,000 cycles
Slip time at cycles, seconds (s) max.	0.74	0.67
Mid-point coefficient, min.	0.097	0.080

4.3.3.6.2 Friction coefficient and wear test. The oil shall be tested in accordance with Caterpillar VC 70 fluid requirements and shall meet the following criteria, as adjusted to accommodate slight changes in individual fluoroelastomer batches and performance of the reference oil:

	Sequence 1220	Sequence FRRET
Avg. dynamic coefficient, %	90 - 140	----
@3000 cycles	----	85 - 130
@8000 cycles	----	90 - 125
@15,000 cycles	----	90 - 125
@25,000 cycles	----	95 - 125
Avg. static coefficient, %	91 - 127	----
Disc wear, mm max.	0.04	----
Energy limit, %	25	----

4.3.3.7 Seal compatibility test. The oil shall be tested in accordance ATD C-4 and shall exhibit results meeting the following nominal criteria, as adjusted to accommodate slight changes in individual elastomer batches:

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Buna, N:	
Volume changes, %	0 to +5
Hardness change, points	0 ± 5
Polyacrylate:	
Volume changes, %	0 to +10
Hardness change, points	0 to +5
Silicone:	
Volume changes, %	0 to +5
Hardness change, points	0 to -10
Fluoroelastomer:	
Volume changes, %	0 to +4
Hardness change, points	-4 to +4
Ethyl Acrylic:	
Volume changes, %	+12 to +28
Hardness change, points	-6 to -18

4.3.3.8 Soot dispersing test. The oil shall be tested in accordance with the ASTM D5967 (T-8) and shall exhibit test results meeting the following criteria:

	1 Test	2 Tests	3 Tests
Viscosity increase, cSt max. from min. corrected to 3.8 % Soot by TGA	11.5	12.5	13.0
Oil consumption grams/brake, horsepower-hour, (g/Bhp-h), max	0.0005	0.0005	0.0005
Filter plugging, differential pressure, (kPa, max.)	138	138	138

4.3.3.9 Corrosion. Complete each test under 4.3.3.9.

4.3.3.9.1 Bearing corrosion. The oil shall be tested in accordance with ASTM D6709 (Seq. VIII) and shall exhibit results meeting the following criteria:

	1 Test	2 Tests	3 Tests
Bearing weight loss, milligrams (mg), avg. max.	29.3	31.9	33.0

4.3.3.9.2 Metal corrosion test. The oil shall be tested in accordance with ASTM D130 and shall not exhibit copper strip discoloration in excess of ASTM No. 3 when compared to ASTM Copper Strip Corrosion Standard, Adjunct to ASTM D130. The oil shall also be tested in accordance with ASTM D5968 and shall exhibit results meeting the following criteria:

Copper, parts per million (ppm) max.	20
Lead, ppm max.	60

4.3.4 Interface requirements verification. Complete each test under 4.3.4.

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4.3.4.1 Homogeneity and miscibility test. An unmixed sample of oil shall show no evidence of separation when tested in accordance with method ASTM D6922, or equivalent (see 4.3.1).

4.3.5 Support and ownership requirement verification. Complete each test under 4.3.5.

4.3.5.1 Hazardous materials test. Using one or more of the methods outlined in 4.3 and 4.3.1 to verify the absence of hazardous materials.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The lubricating oil covered by this specification is intended for the crankcase lubrication of reciprocating compression-ignition engines used in all types of military combat/tactical ground equipment. The oils are also intended for the same application in power transmissions, hydraulic systems, and non-hypoid gear units of engineer/construction equipment, materials handling equipment and combat/tactical ground equipment. The specification product is intended for crankcase lubrication under all conditions of service, as defined by appropriate lubrication orders, when ambient temperatures are in the range shown in Table IV. In addition, the lubricating oil covered by this specification is intended for use in arctic regions as an all-weather (year-round) power transmission fluid for military combat/tactical ground equipment. The lubricating oils covered by this specification meet service classifications API CG-4, CF, and a modified CF-2 of ASTM D4485. The lubricating oil under this specification can be used at all temperatures for transmission, hydraulic, and non-hypoid gear applications provided the use of the arctic grade is warranted for winter operations, otherwise, the appropriate grade under MIL-PRF-2104 is the recommended choice for these applications (see 6.6). This specification requires an oil formulation that is both versatile (used as engine oil, transmission fluid, and hydraulic fluid) and meets extreme operating conditions for use in combat/tactical ground equipment. These requirements cannot be found in current industry standards and are not appropriate for other government vehicles.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of ASSIST to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1).
c. Specification of values and tolerances for finished oil requirements (see Table I and 4.1.1.1).
d. If conformance testing is other than as specified (see 4.1.2).
e. Approved plants or laboratories for testing (see 4.3.2).
f. Packaging requirements, quantity and PIN (see 5 and 1.2).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 46167 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from:

U.S. Army RDECOM-TARDEC
6501 E. 11 Mile Road
ATTN: Ground Systems Performance Team
RDTA-SIE-ES-FPT-GSPF (MS-110)
Warren, MI 48397-5000
usarmy.detroit.rdecom.mbx.tardec-pol-help@mail.mil

An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.4 Conformance testing. DELETED

6.4.1 Other requirements for tolerances for conformance testing. DELETED.

6.5 Reference oils. Reference oils for conducting compatibility tests are to be obtained from the Society of Automotive Engineers (SAE International), 400 Commonwealth Drive, Warrendale, PA 15096, or online at http://www.sae.org.

6.6 Temperature range. Recommended ambient temperature ranges for specific grade oils are shown in Table IV.

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TABLE IV. Recommended ambient temperature ranges for usage of arctic engine oil.

EXPECTED TEMPERATURES													
°F	<-50	-25	0	15	25	40	50	60	70	80	90	100	110	120
°C	<-46	-32	-18	-10	-5	5	10	16	21	27	32	38	44	49
	OEA-30 (O-183) (See Note Below)													
	[redacted]													
	OEA-30 (O-183) Engine													
	[redacted]													

NOTE: For power transmission, hydraulic system and non-hypoid gearbox applications, lubricant may be used at all temperatures.

6.7 Definitions.

6.7.1 Bulk lot. An indefinite quantity of a homogeneous mixture of one grade of oil offered for acceptance in a single, isolated container; or manufactured in a single plant run (not exceeding 24 hours), through the same processing equipment, with no change in the ingredient materials.

6.7.2 Packaged lot. An indefinite number of 208.175 L (55 gallon drum) or smaller unit containers of identical size and type, offered for acceptance, and filled with a homogeneous mixture of one grade of oil from a single, isolated container; or filled with a homogeneous mixture of one grade of oil, manufactured in a single plant run (not exceeding 24 hours), through the same processing equipment, with no change in the ingredient materials.

6.8 Subject term (key word) listing.

Combat service
Compression-ignition
Power transmission fluid
Tactical service
Tribology

6.9 International standardization agreement implementation. This specification implements STANAG 1135 “Interchangeability of Fuels, Lubricants and Associated Products Used by the Armed Forces of the North Atlantic Treaty Nations”. When amendment, revision, or cancellation of this specification is proposed, the preparing activity must coordinate the action with the U.S. National Point of Contact for the international standardization agreement, as identified in the ASSIST database at https://assist.dla.mil.

6.10 Amendment notations. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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6.11 Shelf-life. This specification covers an item where the assignment of a Federal shelf-life code is a consideration. Specific shelf-life requirements should be specified in the contract or purchase order, and should include, as a minimum, shelf-life code, shelf-life package markings in accordance with MIL-STD-129 or FED-STD-123, preparation of a materiel quality storage standard for type II (extendible) shelf-life items, and a minimum of 85 percent shelf-life remaining at time of receipt by the Government. These and other requirements, if necessary, are in DoD 4140.27-M, *Shelf-life Management Manual*. The shelf-life codes are in the Federal Logistics Information system Total Item Record. Additive information for shelf-life management may be obtained from DoD 4140.27-M, or the designated shelf-life Points of Contact (POC). The POC should be contacted in the following order: (1) the Inventory Control Points that manage the item and (2) the DoD Service and Agency administrators for the DoD Shelf-Life Program. Appropriate POCs for the DoD Shelf-Life Program can be contacted through the DoD Shelf-Life Program website: https://www.shelflife.dla.mil/.

Custodians:
Army - AT
Navy - SH
Air Force - 68

Preparing Activity:
Army - AT

(Project 9150-2017-009)

Review Activities:
Army - AR, MI, SM
Navy - AS, MC, OS, SA, YD
Air Force - 11, 184
DLA - GS

Industry Associations:
ASTM
SAE

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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